T H E P E N N S Y L V A N I A R A I L R O A D

REVISION OF PARAGRAPH 2

LOCOMOTIVE MAINTENANCE INSTRUCTIONS NO. L-5-D

METHOD OF APPLYING AND MAINTAINING FLUES AND ARCH PIPES IN LOCOMOTIVE, STATIONARY AND PORTABLE BOILERS AND INSTRUCTIONS FOR WELDING FLUES TO FLUE SHEETS.

Issued Philadelphia, Pa.
September 10, 1941

The following should be substituted for Paragraph No. 2 in the present issue of the Circular:

2. Safe Ends: (a) All safe ending material must be of new seamless steel tubing. For repairing all flues that have a wall thickness of .135" No. 10 B.W.G. or less (tracing size), the thickness of the safe ending material must be .135". For repairing all flues that have a wall thickness of .150" B.W.G. No. 9, or over, the thickness of the safe ending material must be the same as that of the original flue when new.

(b) The outside diameter of safe ending material must be the same as the standard outside diameter of the flue to which it will be applied, except the material used for safe ending the 3-1/2" and 5-1/2" superheater flues, which may be 1/16" larger in diameter to avoid the necessity of opening the end up to suit holes in front flue sheet.

(c) Safe ends must be applied to the firebox end of the flues, except the large superheater flues 3-1/2" and 5-1/2" outside diameter, which must have the safe ends applied to the smokebox end of the flue. The distance from the welds in the flue to the outside of the flue sheet after the flue is applied must not be less than 3-1/2" for the small flues, and 4-1/2" for the superheater flues. Arch pipes and water tubes must not be safe ended.

(d) The weld end of flues and safe ends that are to be welded by the oil fired method must be prepared as shown in Sketch "B". The weld end of flues and safe ends that are to be welded by the electric resistance method must be prepared as shown in Sketch "C". When the wall thickness of the old flues is found to be less than two gauges below that specified on standard tracings for new flues, they must be scrapped. The flues less than 3-1/2" outside diameter must be tested with a load of 9,000 to 10,000 pounds on a tension flue tester under a water pressure of not less than 400 pounds and all the welds must be hammered while under pressure for defects. Flues that are 3-1/2" outside diameter and over must be tested under a water pressure of 400 pounds and all the welds must be hammered while under pressure for defects. The designated mark assigned the welder for identification in cases of failure in service must be stamped on the safe end or ends.